

DRAFT Fractional covering problems on-line on trees

Bartosz Bromblik

TODO

Only important inequalities are those representing optimal vertex covers when the path is of odd length.

First important inequalities are:

$$x + z = \alpha$$

$$Y + 2 \cdot a_0 = 2 \cdot \alpha$$

$$X + Z + 2 \cdot a_1 = 3 \cdot \alpha \quad \equiv \quad 2 \cdot (1 - a_0) + 2 \cdot a_1 = 3 \cdot \alpha$$

$$Y + 2 \cdot (1 - a_1) + 2 \cdot a_2 = 4 \cdot \alpha$$

$$2 \cdot (1 - a_0) + 2 \cdot (1 - a_2) + 2 \cdot a_3 = 5 \cdot \alpha$$

Assumptions

- Every important inequality is treated as equality to make the example easy to solve.
- Symmetry breaking: second edge is connected to the vertex with smaller value (or arbitrarily if $x = y$).

Solution

First let's assume the optimal start of $x = y = \frac{1}{2}$.

Next we need $Y + z = 1$ and $x + z = \alpha$. Therefore $z = \alpha - \frac{1}{2}$ and $y = \frac{3}{2} - \alpha$.

Next we have $2 \cdot a_0 + Y = 2 \cdot \alpha$, from which we get $a_0 = \frac{3}{2}\alpha - \frac{3}{4}$, and $X = Z = 1 - a_0 = \frac{7}{4} - \frac{3}{2}\alpha$.

Next $2 \cdot (1 - a_0) + 2 \cdot a_1 = 3 \cdot \alpha$, therefore $a_1 = 3 \cdot \alpha - \frac{7}{4}$.

Further, we notice a pattern that every i th and $i + 2$ th equations look like:

$$(sth) + 2 \cdot a_k = (k + 2) \cdot \alpha$$

$$(sth) + 2 \cdot (1 - a_{k+1}) + 2 \cdot a_{k+2} = (k + 4) \cdot \alpha$$

We subtract the former from the latter and we get:

$$a_{k+2} + 1 - a_{k+1} - a_k = \alpha \quad \Rightarrow \quad a_{k+2} = a_{k+1} + a_k + (\alpha - 1)$$

Calculating first few terms to notice the pattern:

$$a_0 = 1 \cdot (a_0) + 0 \cdot (a_1) + 0 \cdot (\alpha - 1)$$

$$a_1 = 0 \cdot (a_0) + 1 \cdot (a_1) + 0 \cdot (\alpha - 1)$$

$$a_2 = 1 \cdot (a_0) + 1 \cdot (a_1) + 1 \cdot (\alpha - 1)$$

$$a_3 = 1 \cdot (a_0) + 2 \cdot (a_1) + 2 \cdot (\alpha - 1)$$

$$a_4 = 2 \cdot (a_0) + 3 \cdot (a_1) + 4 \cdot (\alpha - 1)$$

$$a_5 = 3 \cdot (a_0) + 5 \cdot (a_1) + 7 \cdot (\alpha - 1)$$

We can spot the Fibonacci series here, which we define as $F_0 = 0$, $F_1 = 1$, $F_{i+2} = F_{i+1} + F_i$. For $i \geq 1$:

$$a_i = F_{i-1} \cdot a_0 + F_i \cdot a_1 + (F_{i+1} - 1) \cdot (\alpha - 1)$$

Which is obvious if you think about it.

We can now substitute a_0 and a_1 with their respective values:

$$\begin{aligned} a_i &= F_{i-1} \cdot \left(\frac{3}{2}\alpha - \frac{3}{4} \right) + F_i \cdot \left(3 \cdot \alpha - \frac{7}{4} \right) + (F_{i+1} - 1) \cdot (\alpha - 1) \\ &= \frac{1}{4} (\alpha \cdot (6 \cdot F_{i-1} + 12 \cdot F_i + 4 \cdot F_{i+1} - 4) - (3 \cdot F_{i-1} + 7 \cdot F_i + 4 \cdot F_{i+1} - 4)) \\ &= \frac{1}{4} (\alpha \cdot (6 \cdot F_i + 10 \cdot F_{i+1} - 4) - (4 \cdot F_i + 7 \cdot F_{i+1} - 4)) \end{aligned}$$

We assumed that $\forall_i a_i \geq 0$, so:

$$a_i = \frac{1}{4} (\alpha \cdot (6 \cdot F_i + 10 \cdot F_{i+1} - 4) - (4 \cdot F_i + 7 \cdot F_{i+1} - 4)) \geq 0$$

Then:

$$\alpha \geq \frac{4 \cdot F_i + 7 \cdot F_{i+1} - 4}{6 \cdot F_i + 10 \cdot F_{i+1} - 4}$$

We now calculate the limit, knowing that $\lim_{i \rightarrow \infty} \frac{F_{i+1}}{F_i} = \phi = \frac{1+\sqrt{5}}{2}$:

$$\lim_{i \rightarrow \infty} \frac{4 \cdot F_i + 7 \cdot F_{i+1} - 4}{6 \cdot F_i + 10 \cdot F_{i+1} - 4} = \lim_{i \rightarrow \infty} \frac{4 + 7 \cdot \frac{F_{i+1}}{F_i} - \frac{4}{F_i}}{6 + 10 \cdot \frac{F_{i+1}}{F_i} - \frac{4}{F_i}} = \frac{4 + 7 \cdot \phi}{6 + 10 \cdot \phi} = \frac{5 - \sqrt{5}}{4} = \frac{3 - \phi}{2}$$

We again drop the inequality to get $\alpha = \frac{3-\phi}{2}$. From this we calculate:

$$a_0 = \frac{3}{2}\alpha - \frac{3}{4} = \frac{6 - 3 \cdot \phi}{4}$$

$$a_1 = 3 \cdot \alpha - \frac{7}{4} = \frac{11 - 6 \cdot \phi}{4}$$

We now substitute those to our general formula for a_i and get:

$$a_i = F_{i-1} \cdot \frac{6 - 3 \cdot \phi}{4} + F_i \cdot \frac{11 - 6 \cdot \phi}{4} + (F_{i+1} - 1) \cdot \left(\frac{3 - \phi}{2} - 1 \right)$$

...TODO...

$$a_i = \frac{1}{4} (F_{i+6} - \phi \cdot F_{i+5} + 2 \cdot \phi - 2)$$